

PrivML: Test Outputs

Timestamp: 2025-09-19 19:22:26
Model ID: Model 50

Dataset Path: data/Humans
Number of Images in Dataset: 10000
Training Size: 75%
Validation Size: 15%
Testing Size: 5%
Encrypted Testing Subset Size: 5%
Random Seed: 500

Plain Model Training

Epochs: 150
Batch Size: 16
Learning Rate: 0.001
Optimizer: Adamax
Loss Function: MSELoss
Sample Output Count: 3

Plain Model Architecture

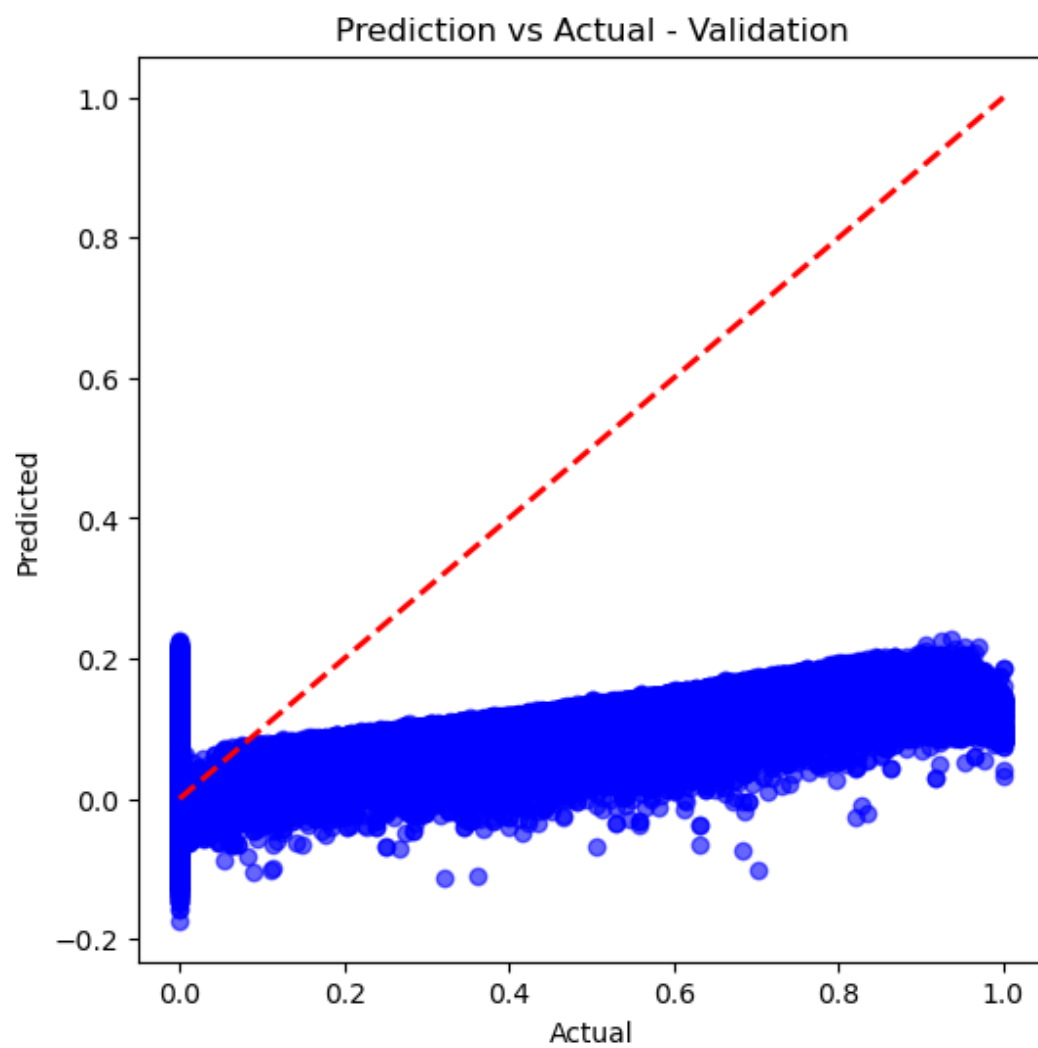
Conv1 Kernel Size: 3
Conv2 Kernel Size: 1
Conv1 Padding: 1

FHE Configuration

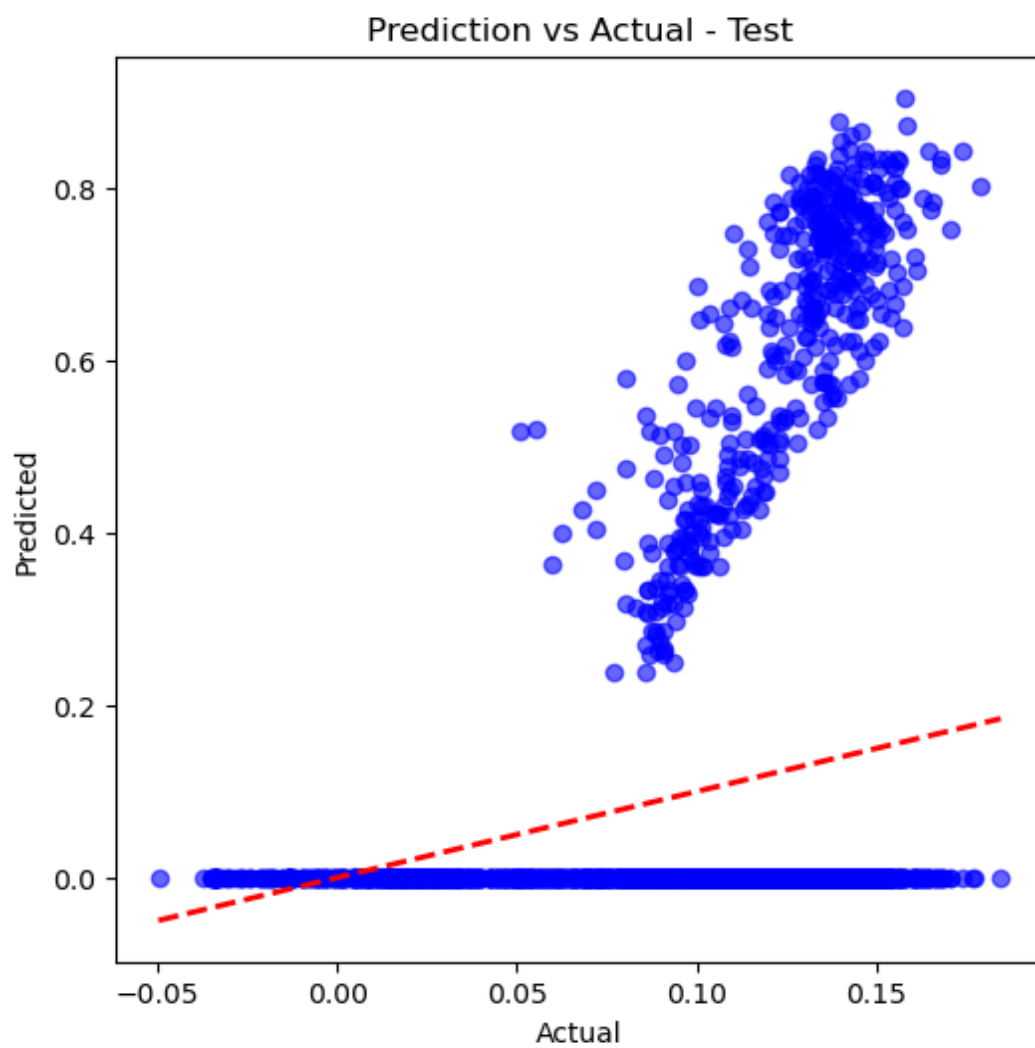
Encryption Scheme: CKKS
Scale (bits): 40
Polynomial Modulus Degree: 32768
Key Generation Level: 60
Decryption Stability Level: 60
Encrypted Loss Function: MSELoss
Coefficient Mod Bit Sizes: [60, 40, 40, 40, 40, 60]

Privacy-Preserving ML: ML Metrics Report

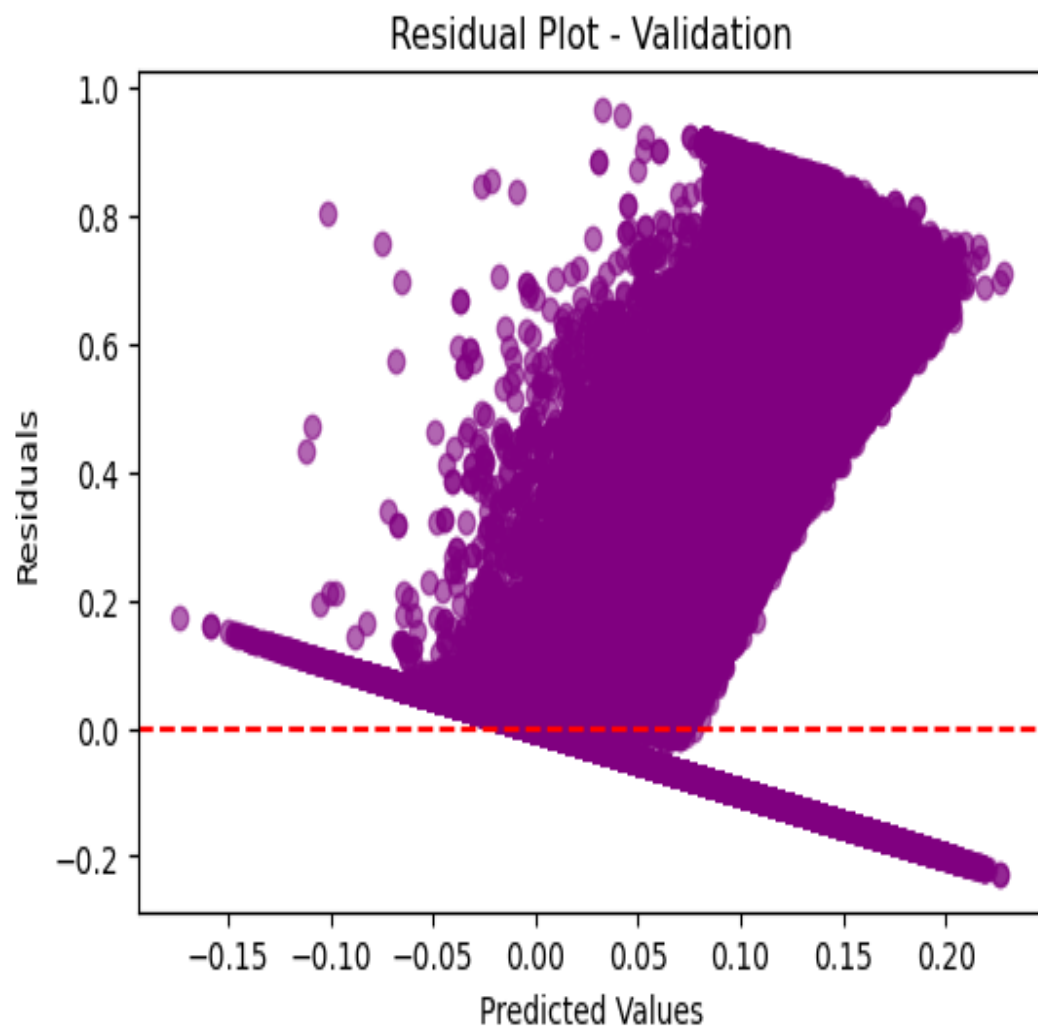
Prediction vs Plot - Validation



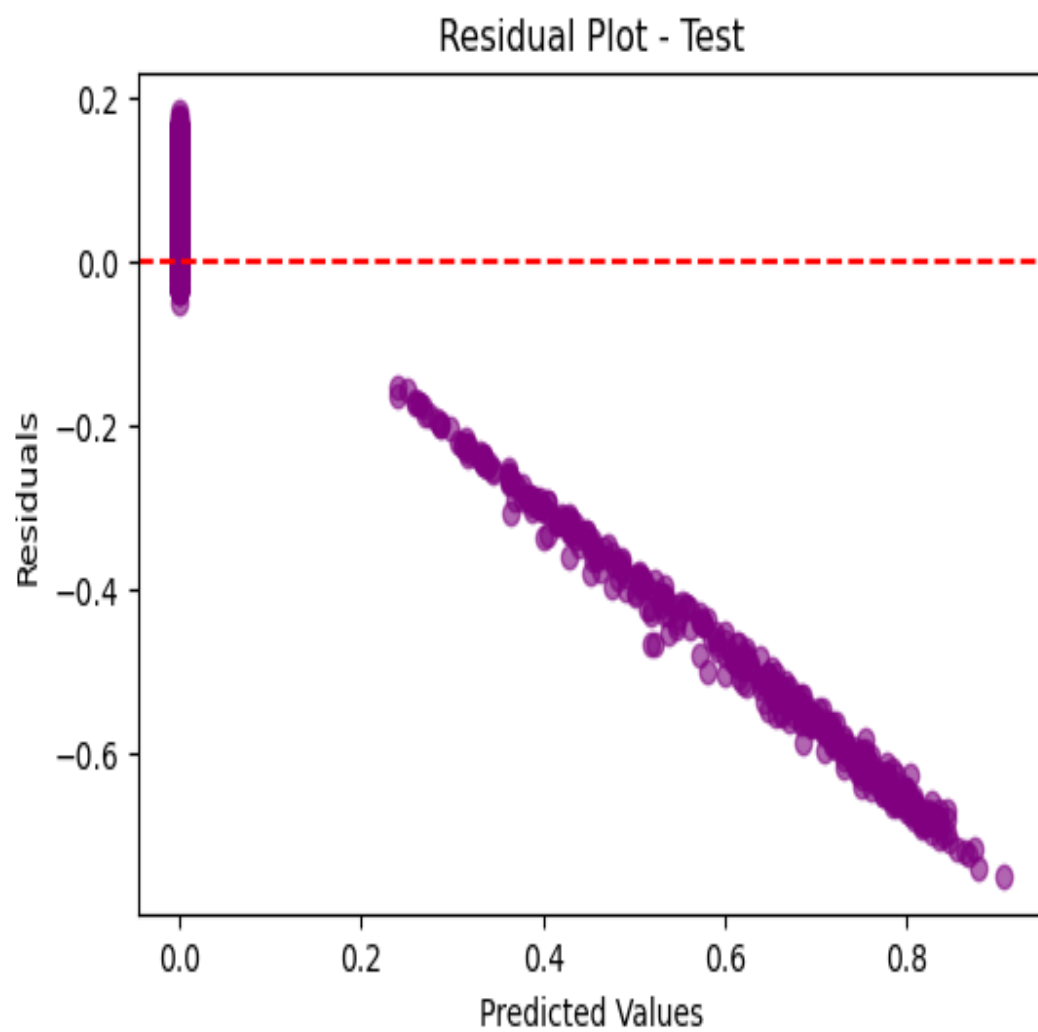
Prediction vs Plot - Test



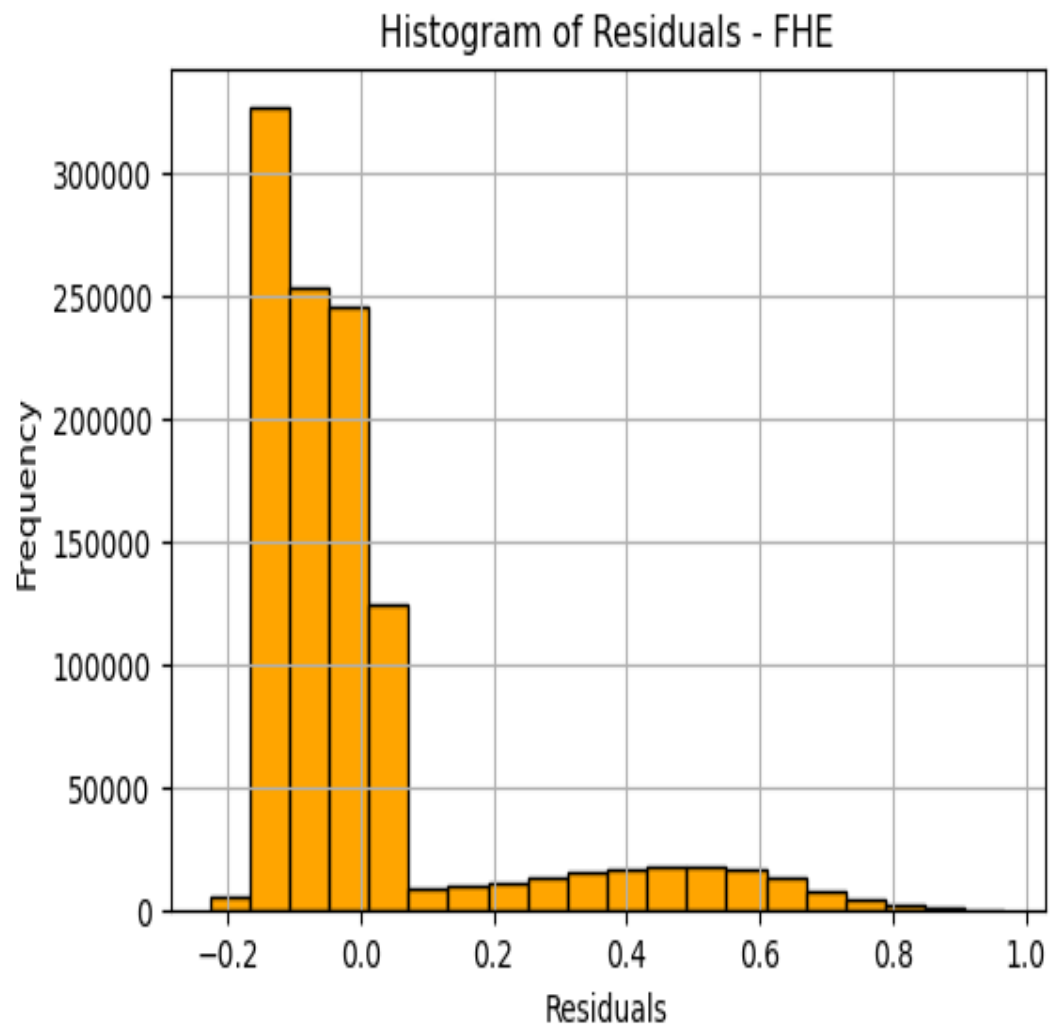
Residual Plot - Validation



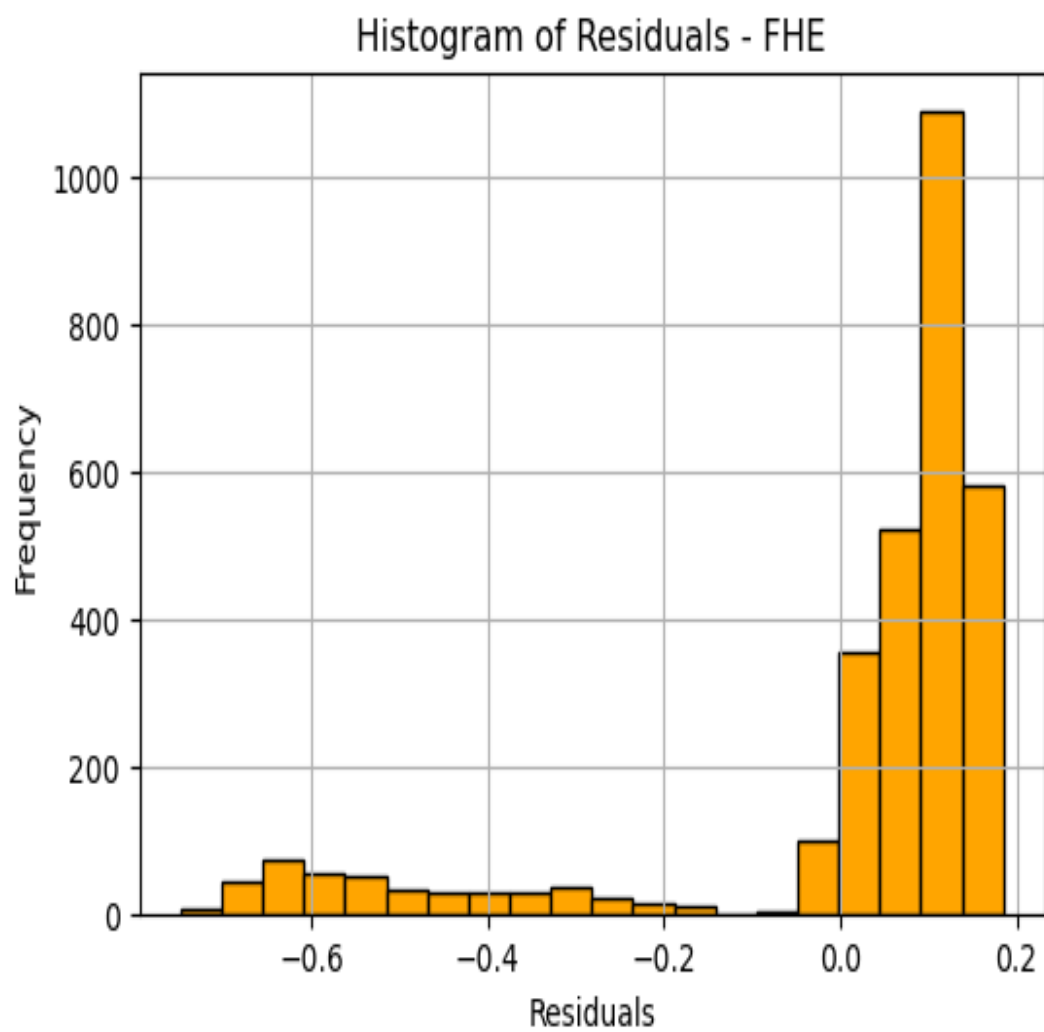
Residual Plot - Test



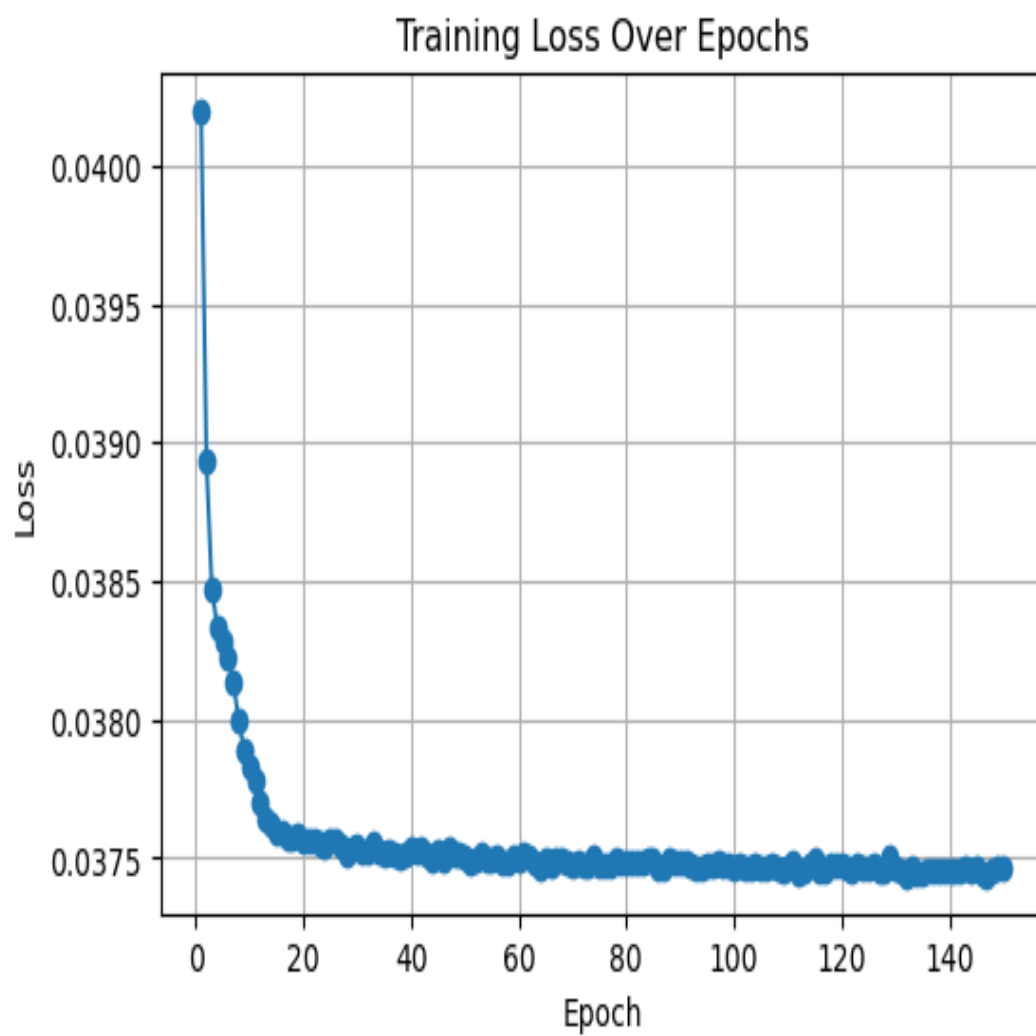
Residual Histogram - Validation



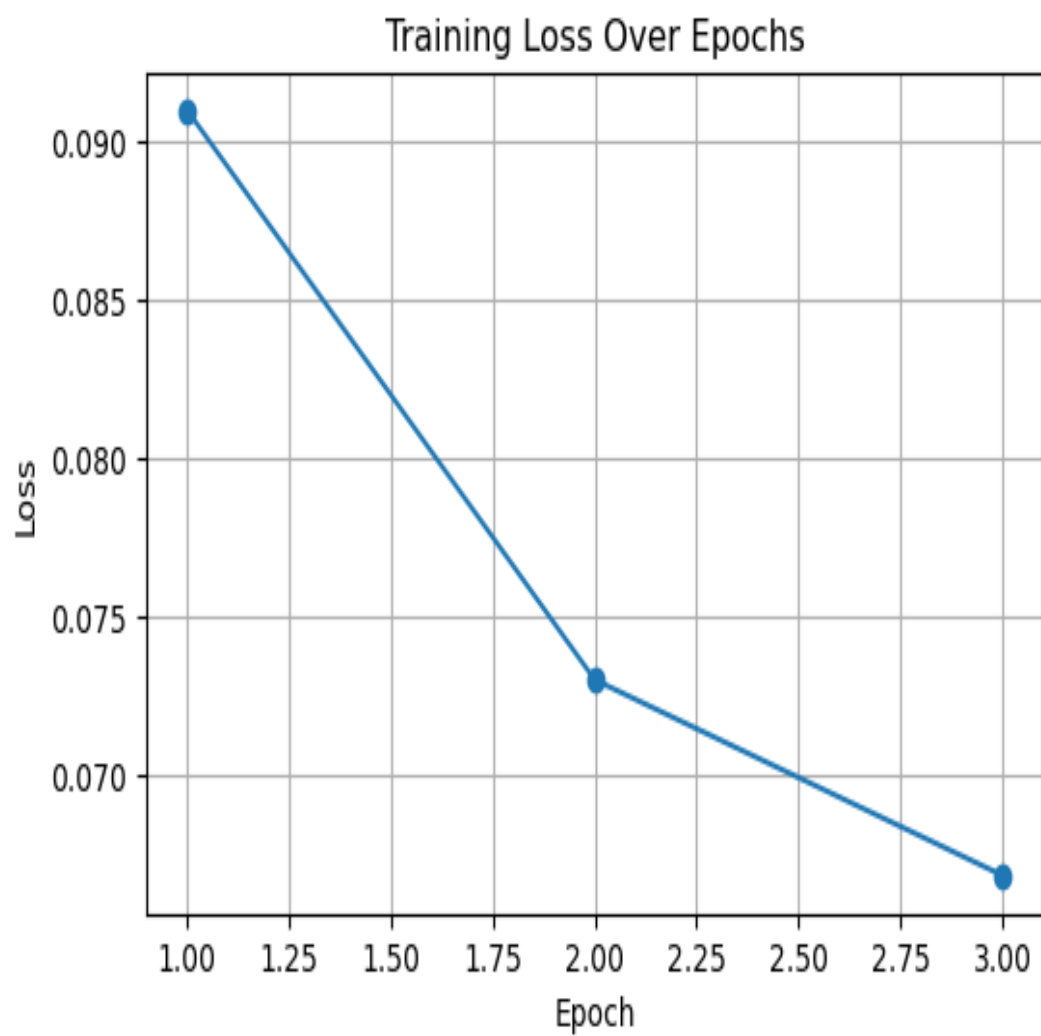
Residual Histogram - Test



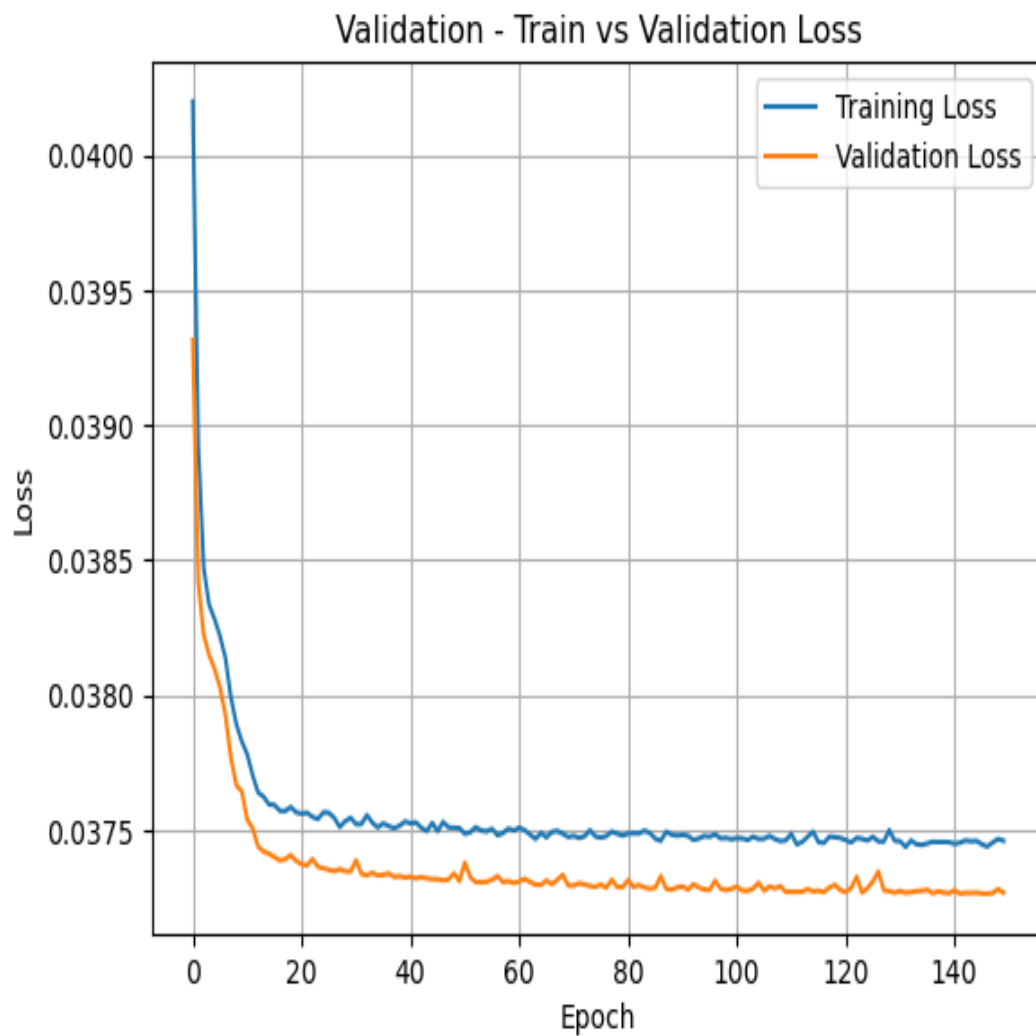
Training Loss - Validation



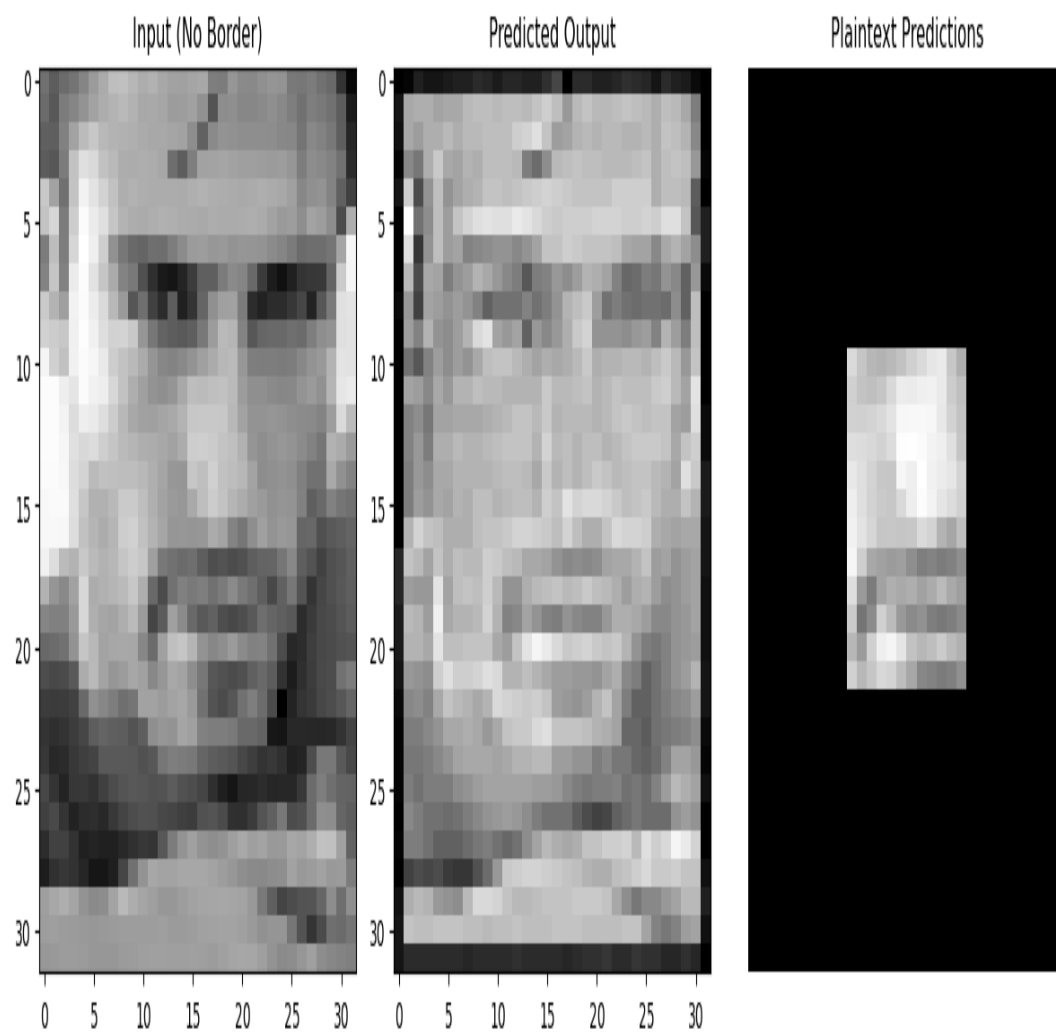
Training Loss - encrypted-fhe



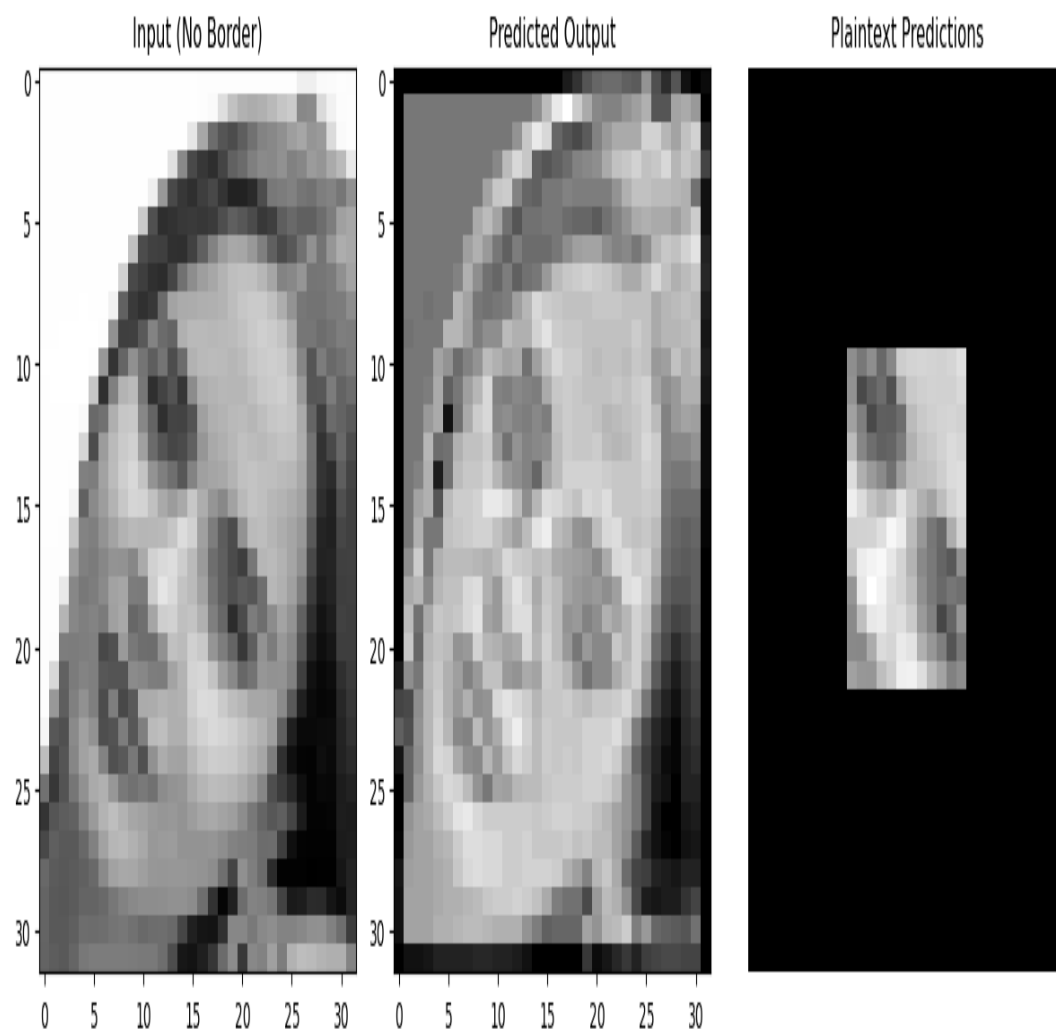
Train vs Validation Loss - Validation



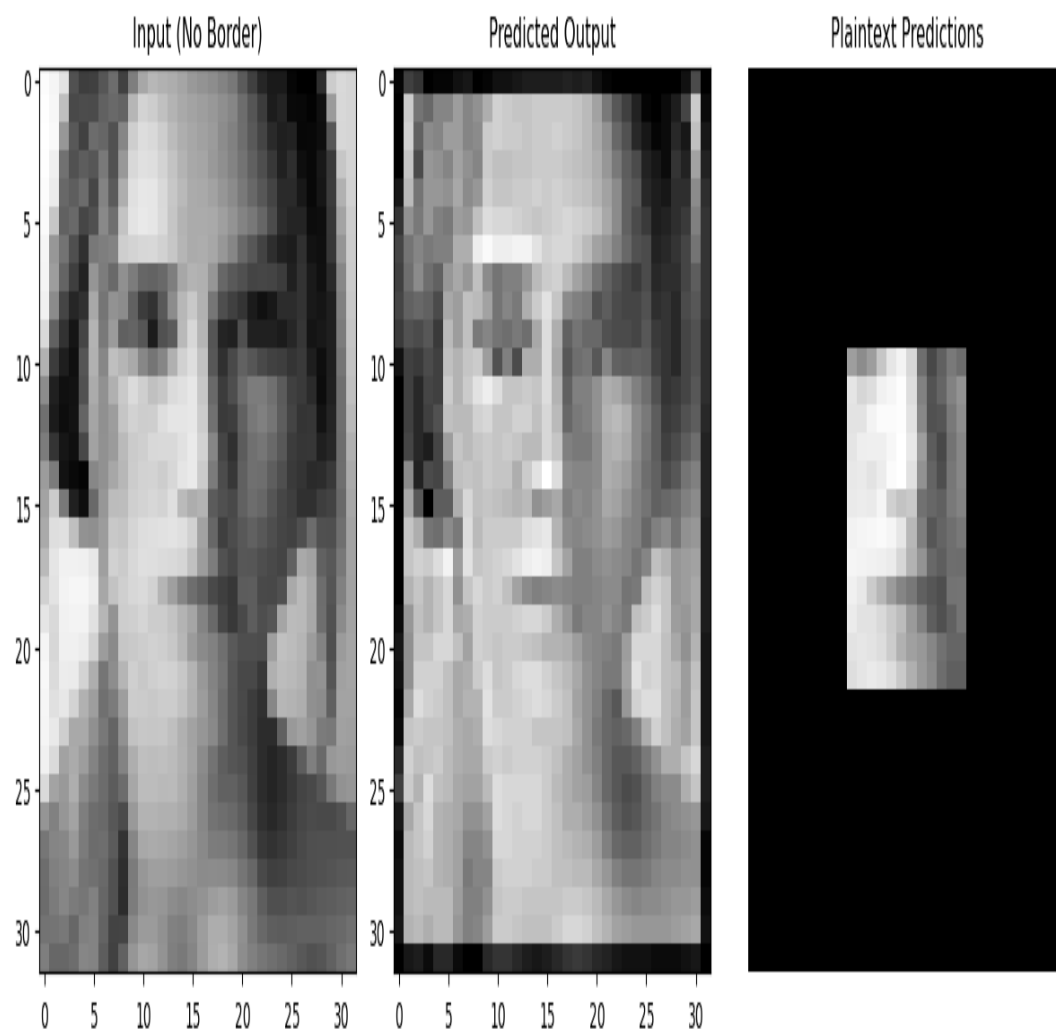
Plaintext: Model Prediction - Plaintext Predicted Image



Plaintext: Model Prediction - Plaintext Predicted Image



Plaintext: Model Prediction - Plaintext Predicted Image



Encrypted: Model Prediction - Encrypted Predicted Image

Encrypted Inference Predictions (Final_Model)

Input Image



Ground Truth



Predicted Output



Input Image



Ground Truth



Predicted Output



Input Image



Ground Truth



Encrypted Predictions



Runtime System Resource Usage

encrypted-fhe: OS: macOS-26.0-arm64-arm-64bit

encrypted-fhe: Python Version: 3.10.8

encrypted-fhe: Processor: arm

encrypted-fhe: CPU Usage (%): 1.6

encrypted-fhe: RAM Usage (Used GB): 6.62

encrypted-fhe: RAM Usage (Total GB): 18.0

encrypted-fhe: RAM Usage (%): 74.3

encrypted-fhe: GPU: Not Available

encrypted-fhe: CPU Usage (%): 1.2

encrypted-fhe: RAM Usage (Used GB): 6.47

encrypted-fhe: RAM Usage (Total GB): 18.0

encrypted-fhe: RAM Usage (%): 73.5

encrypted-fhe: GPU: Not Available

encrypted-fhe: CPU Usage (%): 2.7

encrypted-fhe: RAM Usage (Used GB): 6.41

encrypted-fhe: RAM Usage (Total GB): 18.0

encrypted-fhe: RAM Usage (%): 73.1

encrypted-fhe: GPU: Not Available

